

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Responsible and Safe AI Systems [NPT_4133]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss the major advancements in AI capabilities over the past decade. Identify and explain the imminent risks, such as bias, toxicity, and goal misspecification in modern AI systems. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
B) Apply your understanding of AI risk types to differentiate between imminent and long-term risks in AI systems, using examples such as adversarial attacks and rogue AGI. (5)
Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Illustrate the principle of "Human-Centred and Inclusive AI". Explain how it supports sustainable AI development. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Understand"
B) Examine the mechanisms through which adversarial examples influence the performance of computer vision and language models, and analyze their impacts using relevant examples. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 3) Analyze the explainability techniques such as LIME, SHAP, and Grad-CAM enhance transparency and accountability in AI systems, and evaluate their role in promoting Responsible AI. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Examine the model editing and probing techniques that function to improve the interpretability and transparency of AI models, analyzing their mechanisms and effects. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Evaluate fairness and privacy challenges in AI. Suggest mitigation techniques using fairness metrics or bias reduction tools. (5)
A) Course Outcome: "CO4" / Cognitive Level: "Evaluate"
B) Compare the Digital Personal Data Protection (DPDP) Act, of India and the General Data Protection Regulation (GDPR) of the European Union in promoting Responsible AI. Justify which framework provides stronger safeguards for ethical AI deployment. (5)
Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 5) Critically assess Responsible AI applications in the legal domain, focusing on the roles of transparency and fairness in ensuring ethical and effective outcomes. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- B) Evaluate the potential timeline for achieving AGI and critically propose safeguards to ensure its alignment with human values. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4133:Responsible and Safe AI Systems

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	5
3	3	CO 3	15
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4133:Responsible and Safe AI Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	1	5(10%)
4	Analysis	3	15(30%)
5	Create	4	20(40%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)